ORIGINAL ARTICLE

Tislelizumab Plus Platinum and Etoposide Versus Placebo Plus Platinum and Etoposide as First-Line Treatment for Extensive-Stage SCLC (RATIONALE-312): A Multicenter, Double-Blind, Placebo-Controlled, Randomized, Phase 3 Clinical Trial

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ABSTRACT

Introduction: Extensive-stage SCLC (ES-SCLC) prognosis remains poor. The phase 3 RATIONALE-312 study aimed to evaluate the efficacy and safety of tislelizumab plus chemotherapy as first-line treatment for ES-SCLC.

Methods: RATIONALE-312 is a randomized, double-blind, placebo-controlled trial, conducted in the People's Republic of China. Eligible patients with previously untreated ES-SCLC were randomized 1:1 to receive four cycles of

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tislelizumab 200 mg or placebo, with etoposide plus carboplatin or cisplatin intravenously every 3 weeks, followed by tislelizumab 200 mg or placebo as maintenance. The primary end point was overall survival (OS). Secondary end points included progression-free survival and safety.

Results: Between July 22, 2019 and April 21, 2021, 457 patients were randomized to tislelizumab (n = 227) or placebo (n = 230), plus chemotherapy. Baseline demographics were generally balanced between arms. At the data cutoff (April 19, 2023), the median study follow-up was 14.2 months (interquartile range: 8.6–25.3). Tislelizumab plus chemotherapy exhibited a statistically significant OS benefit versus placebo plus chemotherapy (stratified hazard ratio = 0.75 [95% confidence interval (CI): 0.61–0.93]; one-sided $p = 0.0040$; median: 15.5 [95% CI: 13.5–17.1] versus 13.5 mo [95% CI: 12.1–14.9], respectively). Progression-free survival was significantly improved in the tislelizumab versus placebo arm (stratified hazard ratio = 0.64 [95% CI: 0.52–0.78]; $p < 0.0001$; median: 4.7 [95% CI: 4.3–5.5] versus 4.3 mo [95% CI: 4.2–4.4], respectively). Grade greater than or equal to 3 treatment-related adverse events were reported in 86% of patients in each treatment arm and were mostly hematologic.

Conclusions: Tislelizumab plus chemotherapy exhibited statistically significant clinical benefit and manageable safety compared with placebo plus chemotherapy as first-line treatment in patients with advanced ES-SCLC.

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Keywords: Tislelizumab; Extensive-stage SCLC; Immunotherapy; PD-1

Introduction

SCLC constitutes approximately 15% of all lung cancers,[1,2] and is a highly aggressive disease characterized by early metastasis.[1–3] Most patients (80%–85%) with SCLC are diagnosed with extensive-stage disease (ES-SCLC),[1,2] with exceptionally poor prognosis. The median overall survival (OS) is approximately 10 months, with a 2-year survival rate of less than 15%[2,3] and short progression-free survival (PFS) despite high initial response rates observed with first-line platinum-based chemotherapy regimens.[4,5] These remained the standard of care for several decades until the recent advent of immune checkpoint inhibitors targeting the programmed cell death protein 1 (PD-1) receptor/programmed death-ligand 1 (PD-L1) pathway.[4]

Anti–PD-L1 therapies, when added to platinum plus etoposide as first-line therapy, have consistently exhibited significantly prolonged OS in phase 3 trials in ES-SCLC,[6–10] and evolved into the current first-line standard of care for ES-SCLC.[11] However, no survival benefit was observed in Chinese patients with ES-SCLC enrolled in the extension cohort of the IMpower133 trial (hazard ratio [HR] = 0.927, 95% confidence interval [CI]: 0.600–1.433),[12] suggesting uncertain benefit of combining checkpoint inhibition with chemotherapy in this population. Conflicting results have also been observed in trials of anti–PD-1 therapies when added to platinum plus etoposide in the same setting—whereas the ASTRUM-005 study met its primary end point and found an OS benefit, the KEYNOTE-604 study failed to exhibit a significant improvement in OS ($p = 0.0164$).[13,14] Thus, the potential survival benefits of anti–PD-1 therapies plus chemotherapy as first-line treatment for ES-SCLC remain uncertain.

Tislelizumab is an anti–PD-1 monoclonal antibody that was engineered to minimize Fc gamma receptor binding on macrophages to abrogate antibody-dependent phagocytosis.[15,16] Tislelizumab has exhibited clinical benefit, with an acceptable safety and tolerability profile, in a variety of solid tumors[17–21] and has been approved in the People's Republic of China and Europe.[22,23] In a phase 2 study in a small cohort of patients with ES-SCLC (n = 17), first-line treatment with tislelizumab in combination with platinum plus etoposide resulted in an overall response rate (ORR) of 77%, a median PFS of 6.9 months and a median OS of 15.6 months,[24] supporting further investigation of this combination in this setting.

Here, we report the final analysis results from the phase 3 RATIONALE-312 trial, which evaluated the efficacy and safety of first-line treatment with tislelizumab versus placebo when added to platinum plus etoposide in patients with ES-SCLC.

Materials and Methods

Study Design

RATIONALE-312 was a randomized, double-blind, placebo-controlled, multicenter, phase 3 trial conducted at 51 centers in the People's Republic of China (Supplementary Fig. 1). The trial was conducted in accordance with the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use guidelines, the principles of the Declaration of Helsinki, and country-specific regulatory requirements. The protocol was approved by the relevant institutional review board/independent ethics committee for each study site. An independent data monitoring committee performed regular safety monitoring (at least every 6 mo) and oversaw the interim analysis on OS. A copy of the study protocol is available in the Supplementary Data.

Patients

Patients were eligible for inclusion when they were at least 18 years of age and had histologically or cytologically confirmed ES-SCLC assessed by expert pathologists as defined by the American Joint Committee on Cancer (AJCC) Seventh Edition as Stage IV (T any, N any, M1a or M1b) or T3 to T4 because of multiple lung nodules that were too extensive, or with tumor or nodal volume that was too large to be encompassed in a tolerable irradiation plan. Patients were also required to have an Eastern Cooperative Oncology Group (ECOG) performance status of less than or equal to 1, a life expectancy of at least 12 weeks, adequate organ function, and to have received no previous systemic treatment for ES-SCLC. Patients who were not eligible for inclusion are the following: (1) had active leptomeningeal disease or uncontrolled, untreated brain metastasis; (2) lacked histologic or cytologic ES-SCLC diagnosis or had mixed histologic diagnosis (e.g., SCLC and adenocarcinoma); 3) had underlying medical conditions or alcohol/drug abuse or dependence that would not favor administration of the study drug; or (4) had received previous therapy with an antibody or drug against immune checkpoint pathways (including anti–PD-1, anti–PD-L1, or anti–CTLA-4 antibodies). A full list of inclusion and exclusion criteria can be found in the Supplementary Data. All patients were required to provide written informed consent before participation in the study.

Randomization and Masking

Patients were randomized (1:1) to receive tislelizumab or placebo in a double-blind fashion, with investigators, patients, site staff, and the sponsor blinded to study drug assignment. Randomization was performed using an interactive response technology system by permuted block stratified randomization (block size of four), with stratification factors of ECOG performance status (0 versus 1), investigator-chosen chemotherapy (cisplatin versus carboplatin), and brain metastasis (yes versus no).

Procedures

Induction treatment comprised four 21-day cycles of intravenous tislelizumab 200 mg or placebo every 3 weeks in combination with etoposide (100 mg/m^2 intravenously on days 1–3 of each 21-day cycle) and platinum (investigator's choice of cisplatin [75 mg/m^2] or carboplatin [area under the plasma or serum concentration-time curve = 5]) intravenously on day 1 of each 21-day cycle. Placebo plus chemotherapy was selected as the control arm for this study as this reflected the standard of care for first-line ES-SCLC in the People's Republic of China at the time the study was initiated in mid-2019 (atezolizumab in combination with chemotherapy was approved in the People's Republic of China in February 2020).[25,26] Four cycles of induction treatment were followed by maintenance treatment with either intravenous tislelizumab 200 mg or placebo every 3 weeks according to the patient's assigned treatment arm (Supplementary Fig. 1). Maintenance treatment continued until disease progression, loss of clinical benefit, unacceptable toxicity, or withdrawal of informed consent. Treatment beyond disease progression was permitted at the investigator's discretion, provided protocol-defined criteria for continued treatment were met, as described in the Supplementary Data. Crossover between the two treatment arms was not permitted.

Tumor imaging (computed tomography [CT] scans or magnetic resonance imaging of the chest, abdomen, pelvis, and brain) was performed at baseline, every 6 weeks (±7 days) for the first 48 weeks after cycle 1 day 1, and every 9 weeks (±7 days) thereafter. Tumor responses were assessed by investigators, according to Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST version 1.1). Adverse events (AEs) were recorded for up to 30 days after the last dose of a study drug or until the initiation of another anticancer therapy, whichever occurred first. Any drug-related serious AEs after treatment discontinuation were recorded. Immune-mediated AEs (imAEs) were recorded until 90 days after the last dose of tislelizumab or placebo regardless of whether the patient started a new anticancer therapy. AEs were graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events version 5.0.

Outcomes

The primary end point was OS, defined as the time from randomization until death from any cause. Secondary efficacy end points included investigator-assessed PFS, ORR, duration of response (DoR), disease control rate, safety, and tolerability. The effect of treatment on health-related quality of life was also assessed as a secondary end point and will be reported separately. Assessment of PFS after the next line of treatment (PFS2) was an exploratory end point. A complete list of study end points can be found in the protocol (available in the Supplementary Data).

Statistical Analyses

Efficacy analyses were performed in the intent-to-treat (ITT) population, which included all randomized patients. Safety analyses were performed in the safety analysis set, which included all randomized patients who received at least 1 dose of any study drug, analyzed as actually treated.

The planned sample size of approximately 455 patients was on the basis of the number of events required to exhibit an OS superiority of tislelizumab plus chemotherapy versus placebo plus chemotherapy. For the final analysis of OS, approximately 353 events were required to provide 80% power at a one-sided significance level of 0.025 to detect an HR of 0.74. The median OS for placebo plus chemotherapy was assumed to be 12 months and survival time from both arms was assumed to follow exponential distribution. One interim analysis was planned when approximately 80% of total events occurred, and analysis boundaries were predefined using Lan-DeMets approximation to the O'Brien-Fleming boundary. This article reports the results of the final analysis with a data cutoff of April 19, 2023. The updated *p* value boundary for superiority was 0.0211, which was on the basis of actual events observed at interim analysis and 356 actual events at final analysis.

Once the superiority of OS was exhibited, the full one-sided significance level of 0.025 was passed to the hypothesis testing of PFS. The null hypothesis for OS and PFS was tested using a one-sided *p* value estimated from a log-rank test, stratified by ECOG performance status (1 versus 0) and investigator-chosen chemotherapy (cisplatin versus carboplatin); HRs and 95% CIs were estimated using stratified Cox regression models with treatment arm as a covariate, and the same stratification factors as mentioned for the log-rank test. The median survival time and survival distribution of OS were estimated using the Kaplan-Meier method. The 95% CIs for median OS were calculated using the Brookmeyer and Crowley method. In addition, the cumulative probability of OS at various time points was calculated and presented with two-sided 95% CIs using Greenwood's formula. PFS was analyzed in a similar manner to OS. OS analysis on the basis of the restricted mean survival time method was preplanned as a supplementary analysis to adjust for potential nonproportionality. Multivariate analyses for OS were also conducted, adjusting for key baseline characteristics and prognostic factors on the basis of a stratified Cox regression model, including predefined covariates (liver metastasis, baseline lactate dehydrogenase [LDH], and smoking status), a post hoc added covariate (number of metastatic sites at baseline, identified from observed imbalance), and the previously mentioned stratification factors as strata. Subgroup analyses of OS and PFS were conducted in the following prespecified subgroups: (1) age (<65 versus ≥65 y); (2) sex (male versus female); (3) ECOG performance status (0 versus 1); (4) brain metastases at baseline (yes versus no); (5) choice of platinum chemotherapy (cisplatin versus carboplatin); (6) liver metastases at baseline (yes versus no); (7) baseline LDH (≤ upper limit of normal versus > upper limit of normal); (8) smoking status (never versus current/former); and (9) AJCC stage at study entry (III versus IV). HRs and 95% CIs for the subgroup analyses were estimated using an unstratified Cox regression model.

Confirmation of response was required for analyses on the basis of tumor responses. Patients with no baseline tumor assessment were considered as nonresponders. Clopper-Pearson 95% CIs were estimated for the ORR and disease control rate within each arm. DoR was analyzed in responders using a similar methodology to PFS and a similar methodology was implemented in the PFS2 analysis as well. Descriptive statistics were used for all safety data analyses.

Further details of the statistical analyses can be found in the statistical analysis plan (available in the Supplementary Data). All calculations and analyses were performed using the Statistical Analysis System version 9.4 (SAS Institute, Cary, NC). This study is registered with ClinicalTrials.gov (NCT04005716).

Results

Between July 22, 2019 and April 21, 2021, 457 patients were randomly assigned to receive tislelizumab plus chemotherapy (tislelizumab arm; n = 227) or placebo plus chemotherapy (placebo arm; n = 230) (Fig. 1). Patient demographics and baseline characteristics were generally balanced between treatment arms. The median age was 62 years (interquartile range [IQR] 56–67) in the overall study population; most patients were of male sex (81%), and most had an ECOG performance status of 1 (85%). The baseline tumor burden of patients in the tislelizumab arm was slightly higher than that of patients in the placebo arm, as evidenced by a numerically higher percentage of patients with AJCC stage IV disease, liver metastases, and at least 3 distant metastatic sites (Table 1). Investigator-chosen chemotherapy was carboplatin for 79% of patients and cisplatin for 21% of patients (Fig. 1).

At the final analysis data cutoff (April 19, 2023), the median study follow-up time in the ITT population (from randomization to death, discontinuation from the study, or the data cutoff, whichever occurred first) was 14.2 months (IQR 8.6–25.3); minimum study follow-up time (from last patient randomization to data cutoff) was 23.9 months. Overall, 456 (>99%) randomized patients received any dose of their randomized study treatment. The mean duration of treatment was higher for tislelizumab (39.0 weeks [SD 43.2]) than placebo (23.6 weeks [SD 19.0]), although median duration of treatment was similar (19.4 wk [IQR 15.4–41.1] and 19.1 wk [IQR 17.3–25.7], respectively); more patients in the tislelizumab arm received greater than 16 cycles (≥1 y) of study treatment (Supplementary Table 1). At the final analysis

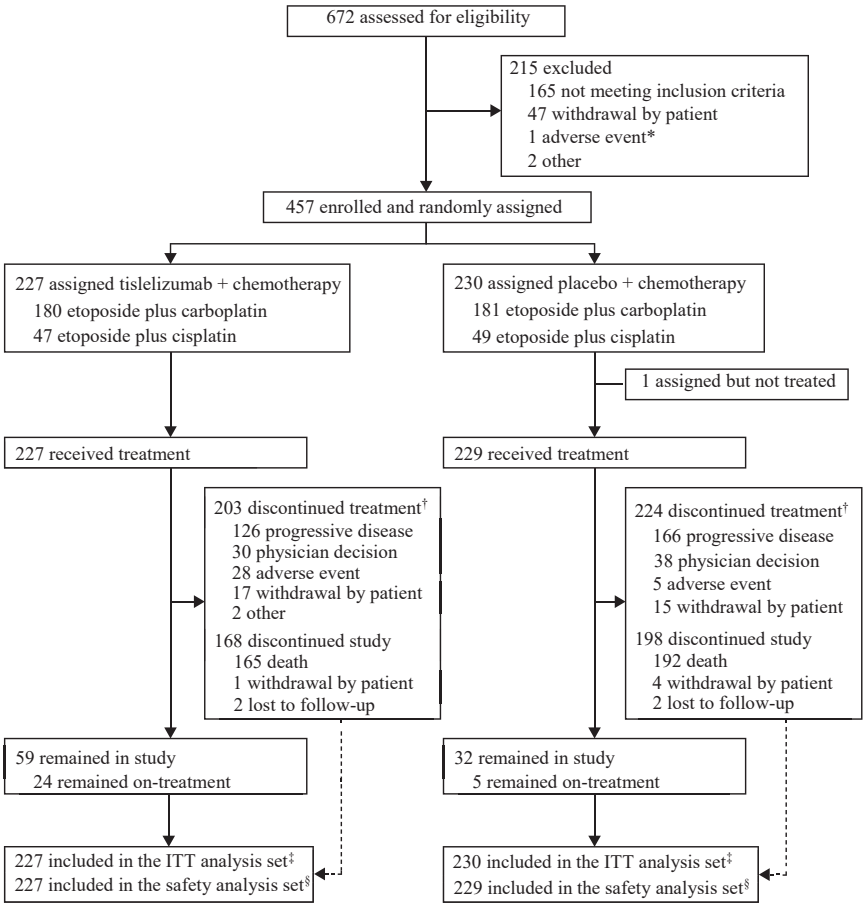


Figure 1. Trial profile. The data cutoff was April 19, 2023. *One patient experienced a serious adverse event of respiratory failure during the screening period. †The stated reason for treatment discontinuation is the reason for discontinuation from the last component. ‡ITT analysis: all randomized patients. §Safety analysis set: all patients who received at least 1 dose of any study drug. ITT, intent-to-treat.

data cutoff (April 19, 2023), tislelizumab treatment was ongoing in 24 of 227 patients (11%) and placebo treatment was ongoing in five of 230 patients (2%). Chemotherapy exposure was similar in the two treatment arms (median of four cycles in both arms) (Supplementary Table 1).

At data cutoff (April 19, 2023), a total of 356 deaths were observed in the ITT population, including 165 of 227 patients (73%) in the tislelizumab arm and 191 of 230 patients (83%) in the placebo arm. The primary study end point was met, with tislelizumab plus chemotherapy exhibiting a statistically significant improvement in OS versus placebo plus chemotherapy in the ITT population (stratified HR = 0.75 [95% CI: 0.61–0.93]; $p = 0.0040$) (Fig. 2*A*). The median OS was 15.5 months (95% CI: 13.5–17.1) in the tislelizumab arm and 13.5 months (95% CI: 12.1–14.9) in the placebo arm. Estimated 1-, 2-, and 3-year OS rates in the tislelizumab versus placebo arms were 63% versus 58%, 33% versus 22%, and 25% versus 9%, respectively. The preplanned restricted mean survival time analysis for OS, with a cutoff at 40.8 months (the minimum of the longest follow-up time in either treatment arm), revealed a 3.2-month difference in OS (95% CI: 0.9–5.5) in favor of tislelizumab plus chemotherapy, indicative of clinical benefit compared with placebo plus chemotherapy. Prespecified multivariate analysis adjusting for liver metastasis, baseline LDH, and smoking status supported the primary analysis, with a stratified HR of 0.72 (95% CI: 0.58–0.89). In addition, a post hoc multivariate analysis adjusting for the observed imbalance in the number of distant metastatic sites as an additional covariate further supported the positive treatment effect in the tislelizumab-contained arm, with a stratified HR of 0.71 (95% CI: 0.57–0.87). An OS benefit of tislelizumab plus chemotherapy versus placebo plus chemotherapy was consistently seen across patient subgroups (Fig. 2*B*).

Tislelizumab plus chemotherapy significantly improved PFS versus placebo plus chemotherapy (PFS events; 176 patients [78%] in the tislelizumab arm versus 207 patients [90%] in the placebo arm; stratified HR = 0.64 [95% CI: 0.52–0.78]; one-sided $p < 0.0001$)

Table 1. Participant Demographics and Baseline Characteristics (ITT Analysis Set)

Demographic/ Characteristic	Tislelizumab Plus Chemotherapy (n = 227)	Placebo Plus Chemotherapy (n = 230)
Age, y		
Median (IQR)	63 (56-66)	62 (56-67)
<65 y	138 (61)	149 (65)
≥65 y	89 (39)	81 (35)
Sex		
Male	186 (82)	186 (81)
Female	41 (18)	44 (19)
ECOG performance status		
0	35 (15)	34 (15)
1	192 (85)	196 (85)
Smoking status		
Never	53 (23)	59 (26)
Current	151 (67)	135 (59)
Former	23 (10)	36 (16)
AJCC stage at study entry[a,b]		
IIIA	4 (2)	2 (1)
IIIB	16 (7)	27 (12)
IV	207 (91)	201 (87)
Number of metastatic sites[c]		
1	2 (<1)	2 (<1)
2	42 (19)	64 (28)
≥3	183 (81)	164 (71)
Liver metastasis		
Yes	64 (28)	59 (26)
No	163 (72)	171 (74)
Brain metastasis		
Yes	1 (<1)	4 (2)
No	226 (>99)	226 (98)
Baseline LDH		
≤ULN	114 (50)	109 (47)
>ULN	113 (50)	121 (53)
Choice of platinum		
Carboplatin	180 (79)	181 (79)
Cisplatin	47 (21)	49 (21)

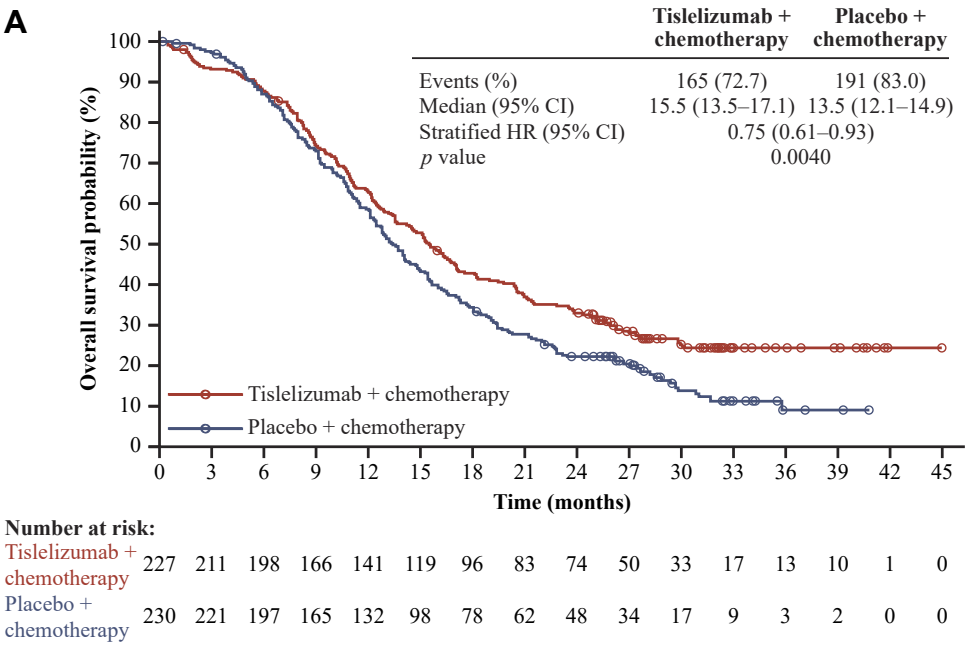
Note: The data cutoff was April 19, 2023. Data are n (%) unless stated otherwise.
[a]Study entry was the date of randomization.
[b]On the basis of AJCC Staging Manual Seventh Edition.
[c]A patient could have multiple metastatic sites.
AJCC, American Joint Committee on Cancer; ECOG, Eastern Cooperative Oncology Group; IQR, interquartile range; ITT, intent-to-treat; LDH, lactate dehydrogenase; ULN, upper limit of normal.

(Fig. 2*C*). The median PFS was 4.7 months (95% CI: 4.3–5.5) in the tislelizumab arm and 4.3 months (95% CI: 4.2–4.4) in the placebo arm. The estimated 6- and 12-month PFS rates in the tislelizumab versus placebo arms were 35% versus 18%, respectively, and 21% versus 5%, respectively. A PFS benefit of tislelizumab plus chemotherapy versus placebo plus chemotherapy was consistently seen across patient subgroups (Supplementary Fig. 2).

The confirmed ORR was numerically greater in the tislelizumab arm (68% [95% CI: 62–74]) versus the placebo arm (62% [95% CI: 55–68]) (Table 2). The median DoR was numerically longer with tislelizumab plus chemotherapy (4.3 months [95% CI: 4.1–5.6]) compared with placebo plus chemotherapy (3.7 months [95% CI: 3.0–4.1]) (Table 2).

A total of 306 patients (67%) in the ITT population received subsequent systemic anticancer therapies after discontinuation of study drugs (136 patients [60%] in the tislelizumab arm; 170 patients [74%] in the placebo arm). The most typically used subsequent systemic anticancer therapies in both arms were conventional chemotherapy (124 patients [55%] in the tislelizumab arm; 155 patients [67%] in the placebo arm), immunotherapy (43 patients [19%] in the tislelizumab arm; 53 patients [23%] in the placebo arm) and targeted therapy (69 patients [30%] in the tislelizumab arm; 73 patients [32%] in the placebo arm), with more patients in the placebo arm receiving multiple lines of subsequent anticancer therapy (Supplementary Table 2). Improved PFS after the next line of treatment after discontinuation of study drugs was observed in the tislelizumab arm (PFS2 stratified HR = 0.71 [95% CI: 0.57–0.89]; median PFS2: 13.1 mo versus 11.0 mo in the tislelizumab and placebo arm, respectively), indicating a sustained benefit with tislelizumab beyond disease progression despite the more frequent use of second-line subsequent anticancer treatment in the placebo arm (Supplementary Fig. 3).

Treatment-related AEs (TRAEs) were reported in 226 of 227 patients (>99%) in the tislelizumab arm and 228 of 229 patients (>99%) in the placebo arm (Table 3). Grade 3 or higher TRAEs occurred in 194 patients (86%) in the tislelizumab arm and 197 patients (86%) in the placebo arm, and were mostly hematologic events including neutropenia (in 127 patients [56%] and 125 patients [55%], respectively), and decreased white blood cell count (in 54 patients [24%] and 63 patients [27%], respectively) (Table 3). TRAEs led to the discontinuation of therapy in 25 patients (11%) and four patients (2%) in the tislelizumab and placebo arms, respectively (Supplementary Table 3). Treatment-related serious AEs were reported in 71 patients (31%) and 41 patients (18%) in the tislelizumab arm and placebo arm, respectively (Supplementary Table 3); all events with an incidence of 2% or more per arm were hematologic. Eight patients (4%) in the tislelizumab arm experienced a TRAE leading to death (Supplementary Table 3), including five events assessed by investigator as related to both tislelizumab and chemotherapy (thrombocytopenia, acute cardiac failure, autoimmune myocarditis, respiratory failure, and death with unknown cause), two assessed as only related to tislelizumab (gastrointestinal



B

Subgroup	Event/total Tislelizumab + chemotherapy	Event/total Placebo + chemotherapy	HR for death (95% CI)	HR (95% CI)
Overall	165/227	191/230		0.75 (0.61–0.93)
Age Group				
<65 years	98/138	122/149		0.76 (0.58–0.99)
≥65 years	67/89	69/81		0.74 (0.53–1.03)
Sex				
Male	136/186	153/186		0.76 (0.61–0.96)
Female	29/41	38/44		0.69 (0.42–1.12)
ECOG performance status				
0	23/35	27/34		0.74 (0.42–1.29)
1	142/192	164/196		0.76 (0.60–0.95)
Smoking status				
Never	36/53	48/59		0.70 (0.45–1.08)
Smoker	129/174	143/171		0.77 (0.60–0.97)
AJCC staging at study entry				
III	11/20	21/29		0.62 (0.30–1.29)
IV	154/207	170/201		0.75 (0.60–0.93)
Liver metastasis				
Yes	54/64	56/59		0.65 (0.44–0.95)
No	111/163	135/171		0.76 (0.59–0.98)
Brain metastasis				
No	164/226	187/226		0.76 (0.61–0.94)
Baseline LDH				
≤ULN	79/114	88/109		0.73 (0.54–0.99)
>ULN	86/113	103/121		0.80 (0.60–1.06)
Choice of platinum				
Cisplatin	35/47	42/49		0.81 (0.51–1.26)
Carboplatin	130/180	149/181		0.74 (0.59–0.94)

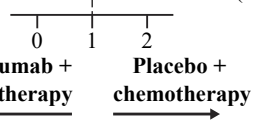


Figure 2. (*A*) Kaplan-Meier plot of OS, (*B*) forest plot of OS by patient subgroups, and (*C*) Kaplan-Meier plot of PFS (ITT analysis set). The data cutoff was April 19, 2023. (*A* and *C*) HRs and 95% CIs for the primary analysis of OS and the analysis of PFS were estimated using a stratified Cox regression model with the treatment arm as a factor and stratified by the actual value of the stratification factors. One-sided *p* values were estimated from log-rank tests stratified by ECOG performance

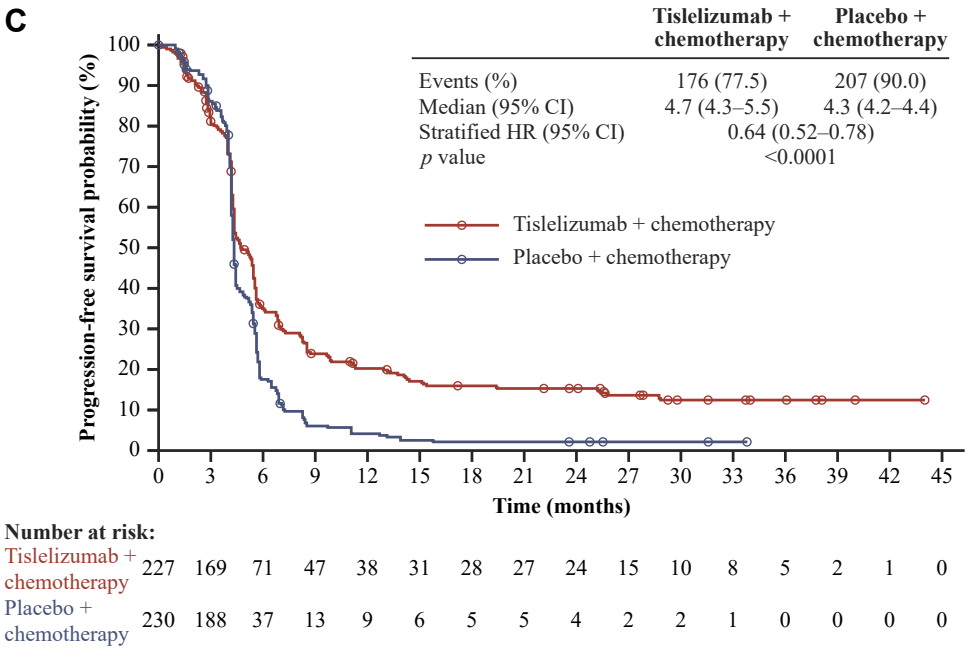


Figure 2. continued

hemorrhage and depressed level of consciousness), and one assessed as only related to chemotherapy (febrile neutropenia). Half of the patients with TRAEs leading to death were older than 60 years old and most had indicators of high tumor burden at baseline (i.e., AJCC stage IV disease with three or more metastatic sites at study entry). No patients in the placebo arm experienced a TRAE leading to death.

ImAEs of any grade were reported in 87 patients (38%) in the tislelizumab arm and 41 patients (18%) in the placebo arm, with hypothyroidism, rash, and hyperthyroidism most typically reported (Table 4 and Supplementary Table 3). Grade 3 or worse imAEs were reported in 25 patients (11%) and one patient (<1%) in the tislelizumab and placebo arms, respectively (Supplementary Table 3). One death because of an imAE was reported in the study (the autoimmune myocarditis case mentioned previously, in the tislelizumab arm).

Discussion

The RATIONALE-312 study met its primary end point, exhibiting superior OS with tislelizumab versus placebo when added to platinum plus etoposide as first-line therapy for ES-SCLC (HR of 0.75 [95% CI: 0.61–0.93]). The OS benefit with tislelizumab plus chemotherapy was consistently observed across all evaluated patient subgroups and was accompanied by a statistically significant improvement in PFS, a greater ORR, and more durable responses versus placebo plus chemotherapy.

As of the data cutoff, the OS data represent a robust analysis with greater maturity, exhibited by a longer follow-up (minimum 23.9 mo), and a higher event-to-patient ratio (78%) than in the previous phase 3 trials reporting OS benefit with the addition of anti–PD-(L)1 treatment to chemotherapy in ES-SCLC,[6–8,10,13,14] acknowledging the caveat of cross-study comparison. Long-lasting treatment benefit was observed in RATIONALE-312 throughout the study follow-up time, as reflected by the higher 2-year survival probability estimated for tislelizumab plus chemotherapy over placebo plus chemotherapy.

The magnitude of OS benefit in RATIONALE-312 was generally comparable with previous studies reporting significant improvements in OS with the addition of anti–PD-1/PD-L1 therapies to first-line chemotherapy in ES-SCLC (IMpower133, CASPIAN, CAPSTONE-1, and ASTRUM-005),[6–8,13] despite the outperformance of the median OS in the control arm of RATIONALE-312 compared with the comparator arms in similar studies

status and investigator-chosen chemotherapy. The median OS follow-up time was estimated by the reverse Kaplan-Meier method with 95% CIs estimated using the Brookmeyer and Crowley method. (*B*) HRs and 95% CIs for the subgroup analysis of OS were estimated using an unstratified Cox regression model. AJCC, American Joint Committee on Cancer; CI, confidence interval; ECOG, Eastern Cooperative Oncology Group; HR, hazard ratio; ITT, intent-to-treat; LDH, lactate dehydrogenase; OS, overall survival; PFS, progression-free survival; ULN, upper limit of normal.

Table 2. Tumor Response Outcomes (ITT Analysis Set)

Outcome	Tislelizumab Plus Chemotherapy (n = 227)	Placebo Plus Chemotherapy (n = 230)
ORR [95% CI]	155 (68%) [62-74]	142 (62%) [55-68]
Best overall response, n (%)		
CR	1 (<1)	0 (0)
PR	154 (68)	142 (62)
Stable disease	46 (20)	60 (26)
Non-CR/Non-PD	0 (0)	1 (<1)
PD	11 (5)	15 (7)
NE[a]/NA[b]	15 (7)	12 (5)
DCR, n (%) [95% CI]	201 (89) [84-92]	203 (88) [83-92]
Median DoR, months [95% CI]	4.3 [4.1-5.6]	3.7 [3.0-4.1]

Note: The data cutoff was April 19, 2023.
[a]Patients who had at least 1 postbaseline tumor assessment, none of which were evaluable for response determination (e.g., not all target lesions captured) on the basis of RECIST v1.1.
[b]Patients with no postbaseline tumor assessment by the data cutoff, including those who discontinued the study (any reason) or died without having any postbaseline tumor assessment.
CI, confidence interval; CR, complete response; DCR, disease control rate; DoR, duration of response; ITT, intent-to-treat; NA, not available; NE, not evaluable; ORR, overall response rate; PD, progressive disease; PR, partial response; RECIST v1.1, Response Evaluation Criteria in Solid Tumors version 1.1.

(13.5 mo versus 10.3–12.8 mo, respectively),[6–8,13] with the caveat of cross-trial comparison. The better-than-expected performance in the control arm of RATIONALE-312 may be partially explained by the higher utilization of subsequent anticancer treatment (mostly anlotinib [30% of patients in the placebo arm], which has been found to improve PFS and OS as third-line or later targeted therapy in Chinese patients with SCLC, and other anti–PD-(L)-1 drugs [23% of patients in the placebo arm]) compared with other studies.[6,8,27] The

Table 3. Treatment-Related AEs by Preferred Term (Safety Analysis Set)

	Tislelizumab Plus Chemotherapy (n = 227)		Placebo Plus Chemotherapy (n = 229)	
Preferred Term	Any grade	Grade ≥3	Any grade	Grade ≥3
Any	226 (>99)	194 (85)	228 (>99)	197 (86)
Alopecia	177 (78)	0 (0)	182 (79)	4 (2)
Anemia	172 (76)	37 (16)	180 (79)	38 (17)
Neutropenia	156 (69)	127 (56)	161 (70)	125 (55)
WBC count decreased	126 (56)	54 (24)	148 (65)	63 (27)
Thrombocytopenia	112 (49)	44 (19)	119 (52)	58 (25)
Nausea	92 (41)	1 (<1)	93 (41)	1 (<1)
Leukopenia	75 (33)	26 (11)	56 (24)	24 (10)
Decreased appetite	73 (32)	1 (<1)	76 (33)	0 (0)
ALT increased	60 (26)	3 (1)	40 (17)	1 (<1)
Constipation	50 (22)	0 (0)	27 (12)	0 (0)
Neutrophil count decreased	50 (22)	33 (14)	56 (24)	49 (21)
Vomiting	48 (21)	1 (<1)	48 (21)	0 (0)
AST increased	43 (19)	3 (1)	31 (13)	0 (0)
Platelet count decreased	30 (13)	8 (3)	34 (15)	8 (3)
Hypothyroidism	29 (13)	1 (<1)	9 (4)	0 (0)
Rash	28 (12)	5 (2)	18 (8)	0 (0)
Malaise	26 (11)	4 (2)	24 (10)	0 (0)
Hypoalbuminemia	24 (11)	0 (0)	17 (7)	0 (0)
Fatigue	23 (10)	3 (1)	12 (5)	0 (0)
Hyponatremia	23 (10)	4 (2)	14 (6)	3 (1)
Lymphocyte count decreased	23 (10)	7 (3)	23 (10)	8 (3)

Note: The data cutoff was April 19, 2023. Data are number of patients (% of patients). TRAEs with an incidence of at least 10% of patients (either treatment arm) are reported. AEs were graded on the basis of NCI CTCAE version 5.0 and coded using MedDRA version 24.0.
AE, adverse event; ALT, alanine aminotransferase; AST, aspartate aminotransferase; MedDRA, Medical Dictionary for Regulatory Activities; NCI CTCAE, National Cancer Institute Common Terminology Criteria for Adverse Events; TRAE, treatment-related adverse event; WBC, white blood cell.

Table 4. Most Common (≥1% of Patients in Either Treatment Arm) Immune-Mediated Adverse Events (Safety Analysis Set)

	Tislelizumab Plus Chemotherapy (n = 227)		Placebo Plus Chemotherapy (n = 229)	
Preferred Term	Grade		Grade	
Patients, n (%)	Any	≥3	Any	≥3
Hypothyroidism	31 (14)	1 (<1)	9 (4)	0 (0)
Rash	30 (13)	5 (2)	18 (8)	0 (0)
Hyperthyroidism	13 (6)	0 (0)	2 (<1)	0 (0)
Pneumonitis	10 (4)	4 (2)	5 (2)	0 (0)
Diabetes mellitus	4 (2)	3 (1)	0 (0)	0 (0)
Immune-mediated myocarditis	4 (2)	0 (0)	0 (0)	0 (0)
Immune-mediated hepatitis	3 (1)	2 (<1)	0 (0)	0 (0)
Immune-mediated lung disease	3 (1)	1 (<1)	1 (<1)	0 (0)
Myositis	3 (1)	0 (0)	1 (<1)	0 (0)
Thyroxine free decreased	3 (1)	0 (0)	0 (0)	0 (0)
Thyroxine increased	1 (<1)	0 (0)	4 (2)	0 (0)

Note: The data cutoff was April 19, 2023. Adverse events were coded using MedDRA version 24.0 and graded for severity using CTCAE version 5.0.
CTCAE, Common Terminology Criteria for Adverse Events; MedDRA, Medical Dictionary for Regulatory Activities.

differences in subsequent therapy across studies may pertain to different clinical practices and drug availability for further lines of treatment for ES-SCLC because of the geographic variation between studies (e.g., immunotherapy [23% versus 5%–7%] and targeted therapy [32% versus <1%] were more frequently used in the control arm of RATIONALE-312 than in the control arm of the global IMpower133 and CASPIAN studies, respectively).[6,8] Whereas the kinase inhibitor anlotinib has been approved as a subsequent targeted therapy for SCLC in the People's Republic of China only,[27] the OS benefit in RATIONALE-312 was sustained in the tislelizumab plus chemotherapy arm and was accompanied by improved PFS2 in favor of the tislelizumab arm, exhibiting the advantages of this therapy over chemotherapy in the first-line setting and regardless of subsequent anticancer treatment.

The patient population enrolled in the RATIONALE-312 study was broadly representative of patients with untreated ES-SCLC in the People's Republic of China. SCLC primarily occurs in smokers,[2,28,29] and the proportion of cases occurring in never-smokers has been reported to be only 2% to 5%.[29,30] In the RATIONALE-312 study, the proportion of patients who were never-smokers (24.5%) was higher than global estimates (∼2.5%),[30,31] consistent with the observed higher prevalence of never-smoking SCLC patients in Asia,[28,29,32–34] and with previous phase 3 trials conducted in the People's Republic of China in the same disease setting (20% in ASTRUM-005; 22% in CAPSTONE-1).[7,13] The high incidence of never-smokers could be attributed to regional differences, whereas environmental (passive) tobacco smoking has been found to contribute to the development of SCLC in never-smokers.[35] The genomic profile of never-smoking patients needs to be explored; however, because of limited sample collection, these data are not available for the RATIONALE-312 study. Despite the regional differences in the incidence of never-smokers, a consistent clinical benefit for OS and PFS was seen with tislelizumab plus chemotherapy versus placebo plus chemotherapy, regardless of smoking status in the RATIONALE-312 study.

Carboplatin or cisplatin are recommended as the platinum component of first-line ES-SCLC regimens, in combination with etoposide.[4,36] However, most ES-SCLC phase 3 trials exhibiting an OS benefit with the addition of anti–PD-1 or anti–PD-L1 therapies to chemotherapy have been restricted to carboplatin-based regimens,[7,8,13] with the exception of the CASPIAN study of the anti–PD-L1 therapy durvalumab, which allowed investigators to choose carboplatin or cisplatin.[6] The phase 3 KEYNOTE-604 study also allowed the use of both platinum regimens; however, the study failed to exhibit a significant OS benefit with pembrolizumab plus chemotherapy as a first-line treatment for ES-SCLC (median OS: 10.8 mo in the pembrolizumab arm versus 9.7 mo in the placebo arm; HR: 0.80; 95% CI 0.64–0.98).[14] Therefore, to our knowledge, RATIONALE-312 is the first phase 3 clinical trial demonstrating an OS benefit with an anti–PD-1 therapy combined with either cisplatin or carboplatin plus etoposide in this setting, providing new evidence that could broaden treatment options for patients with ES-SCLC.

Tislelizumab plus chemotherapy is well tolerated with a manageable safety profile, as approximately one-fifth of the patients received tislelizumab for more than 1 year, whereas few patients discontinued study treatment because of TRAEs. The most typically reported TRAEs occurred in similar proportions of patients in the

two arms and were hematologic in nature, consistent with the known toxicities of platinum-based chemotherapy.[4] The safety profile observed with tislelizumab plus chemotherapy was in line with that reported for anti–PD-1/PD-L1 therapies in combination with chemotherapy in ES-SCLC.[6–8,13,14]

This study has a couple of limitations. First, all patients were enrolled from the People's Republic of China and chemotherapy was selected as the control arm because immunotherapy had not been approved for ES-SCLC in the People's Republic of China before the initiation of the study. Despite the regional representation, the study results are consistent with those from previous trials conducted in other regions[6,10] and were supportive of the concept of chemoimmunotherapy in worldwide clinical practice. Second, only a few patients with brain metastases at baseline were enrolled (1%). The low prevalence of brain metastases at enrollment was similar to that observed in CAPSTONE-1 (2%),[7] which is aligned with regional medical practice in the People's Republic of China for patients with ES-SCLC who seldom receive palliative radiotherapy for their asymptomatic central nervous system lesions. Whereas the survival benefit observed with other PD-L1 inhibitors in combination with chemotherapy has been found to be independent of the presence of baseline brain or central nervous system metastases,[37] the effect of tislelizumab plus chemotherapy in patients with brain metastases at baseline remains to be investigated. In addition, the correlation between PD-L1 expression and efficacy was not included in this report. However, previous trials have provided evidence suggesting that PD-L1 expression is not a predictive biomarker of response to anti–PD-1/PD-L1 treatment in this setting.[7,10,13] Therefore, exploration of potential biomarkers beyond PD-L1 expression is warranted.

In conclusion, the addition of tislelizumab to standard first-line chemotherapy with platinum and etoposide exhibited a significant reduction in the risk of death for patients with ES-SCLC, which was further supported by improvements in PFS, ORR, and DoR, and a manageable safety profile. These results support tislelizumab plus chemotherapy as a potential new first-line treatment option for patients with ES-SCLC.

CRediT Authorship Contribution Statement

Ying Cheng: conceptualization; data curation; writing – original draft, writing – review & editing, visualization.

Yun Fan: conceptualization; data curation; writing – review & editing.

Yanqiu Zhao: data curation; writing – review & editing.

Dingzhi Huang: conceptualization; data curation; writing – review & editing.

Xingya Li: data curation; writing – review & editing.

Peng Zhang: data curation; writing – review & editing.

Mafei Kang: data curation; writing – review & editing.

Nong Yang: data curation; writing – review & editing.

Diansheng Zhong: data curation; writing – review & editing.

Zhen Wang: data curation; writing – review & editing.

Yan Yu: data curation; writing – review & editing.

Yu Zhang: data curation; writing – review & editing.

Jun Zhao: conceptualization; data curation; writing – review & editing.

Tai Qin: conceptualization; data curation; writing – original draft; writing – review & editing, validation, visualization; project administration.

Chenqi Chen: conceptualization; data curation; writing – original draft; writing – review & editing, validation, visualization.

Shiangjiin Leaw: conceptualization; data curation; writing – original draft; writing – review & editing.

Wenjuan Zheng: conceptualization; data curation; writing – original draft; writing – review & editing.

Yong Song: conceptualization; data curation; writing – original draft, writing – review & editing.

Disclosure

Dr. Zhao reports receiving payment or honoraria for lectures, presentations, speaker bureaus, manuscript writing, or educational events from Roche, AstraZeneca, Merck Sharp & Dohme, Novartis, and Eli Lilly. Drs. Qin and Zheng are employees of BeiGene (Beijing) Co., Ltd. and have stock or stock options with BeiGene (Beijing) Co., Ltd. Dr. Leaw and Ms. Chen are employees of BeiGene (Shanghai) Co., Ltd. and have stock or stock options with BeiGene (Shanghai) Co., Ltd. The remaining authors declare no conflict of interest.

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Data Sharing

On request, and subject to certain criteria, conditions, and exceptions, BeiGene, Ltd., will provide access to individual de-identified participant data from BeiGene-sponsored global interventional clinical studies conducted for medicines (1) for indications that have been approved or (2) in programs that have been terminated. BeiGene will also consider requests for the protocol, data dictionary, and statistical analysis plan. Data requests may be submitted to DataDisclosure@beigene.com

Ethics Committee Approval and Informed Consent

All relevant institutional review boards/independent ethics committees approved the study, which was carried out in accordance with the International Conference on Harmonisation Good Clinical Practice Guideline, the principles of the Declaration of Helsinki, and local laws and regulations. All patients provided written informed consent before participation. An independent data monitoring committee assessed safety and efficacy during the study.

Supplementary Data

Note: To access the supplementary material accompanying this article, visit the online version of the *Journal of Thoracic Oncology* at www.jto.org and at [comp inserts doi link here https://doi.org/10.1016/j.jtho.2024.03.008].

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